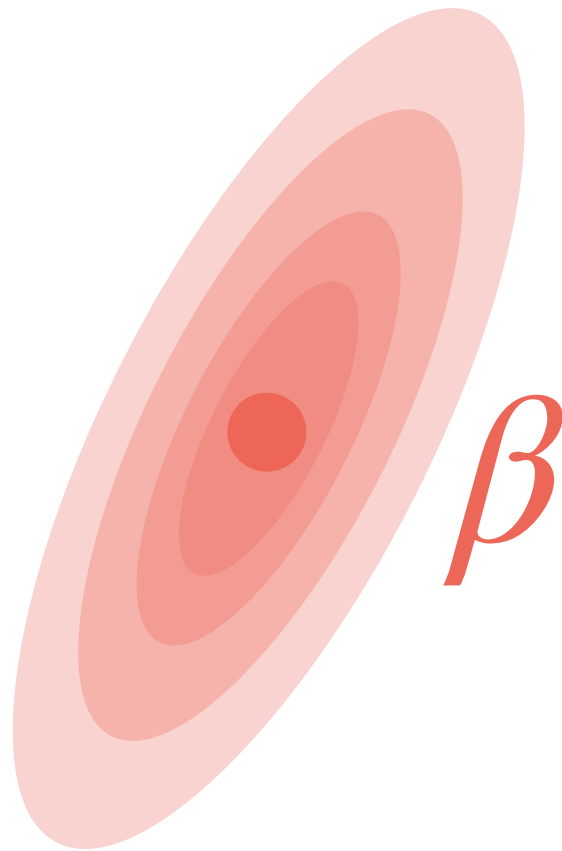


loss function that measures fidelity to β



Parametric Distribution



Target Distribution

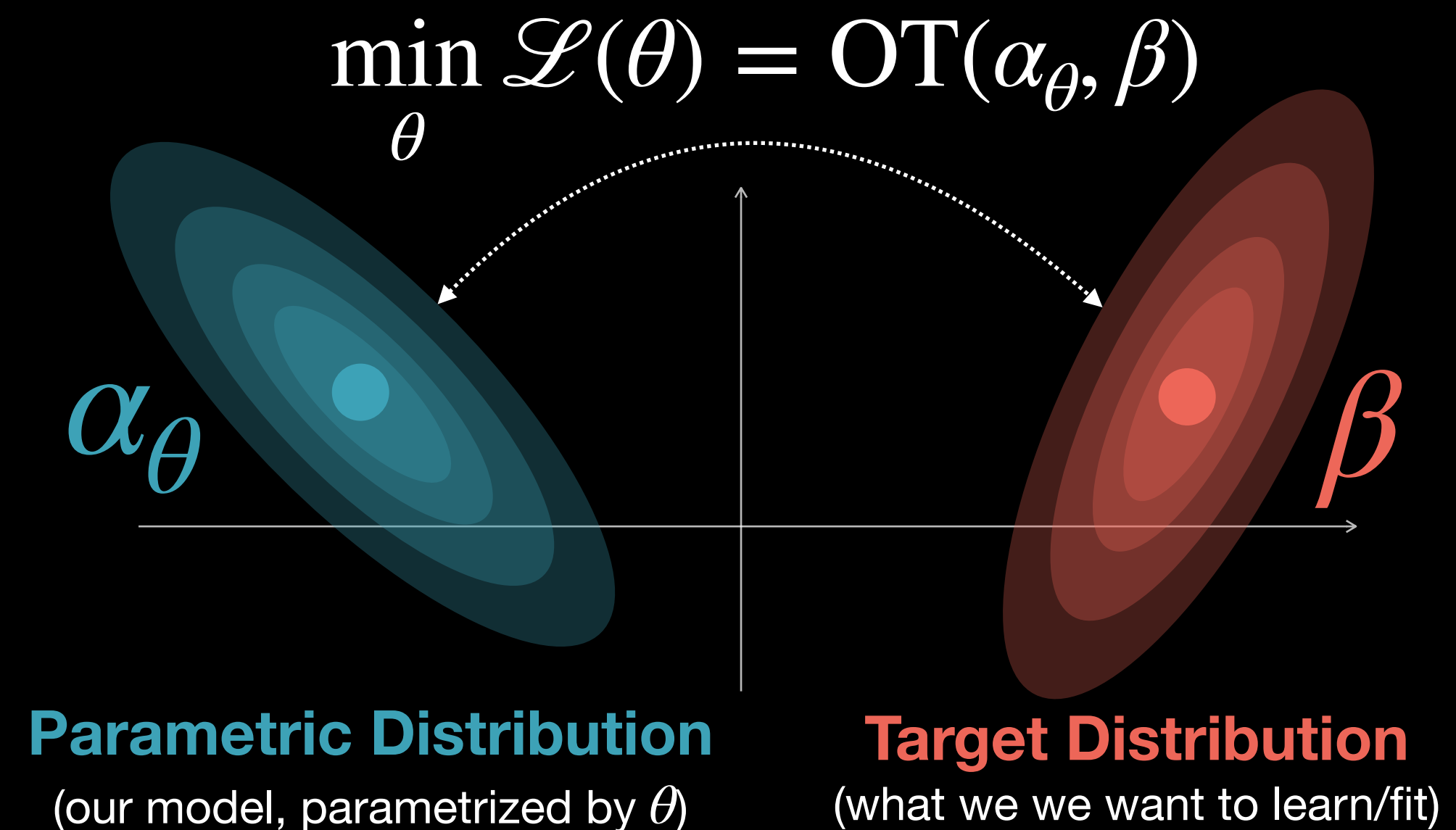
OT as a Variational Objective

- Common case: OT as a loss function to learn / fit a distribution
- Setup: want to model true data distribution β with α_θ from parametric model class $\{\alpha_\theta, \theta \in \Theta\}$
- Examples:
 - For empirical measures, θ can encode locations $\mathbf{x}_i$ and/or weights a_i directly, or the parameters of a prior that generates them
 - For continuous measures, θ can be parameters of map (eg Neural Net) and take $\alpha_\theta = (g_\theta)_\# \zeta$ for some (simple) base measure ζ (e.g. Gaussian)

- Given **loss function that measures fidelity to β** , objective is:

$$\min_{\theta \in \Theta} \mathcal{L}(\theta) = \mathcal{L}(\alpha_\theta; \beta)$$

- Here, we are interested in: $\mathcal{L}(\alpha_\theta, \beta) = \text{OT}_c(\alpha_\theta, \beta)$
- Same setup extends to barycenters, dictionary learning, etc



What Makes a Good ML Loss Function?